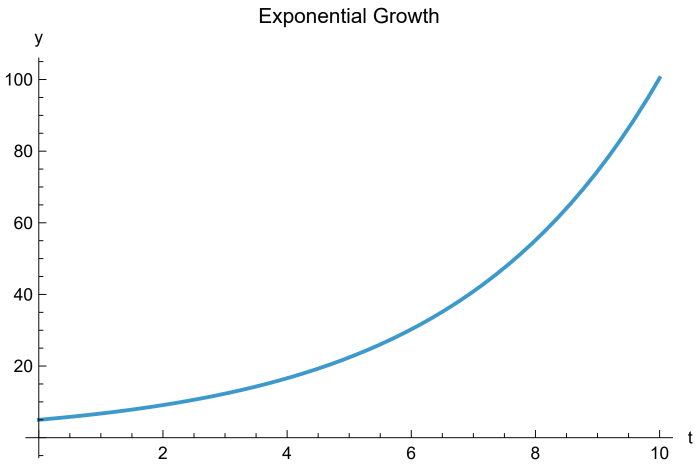
In[◦]:=
```
k = 0.3;

sol = DSolve[{y'[t] == k y[t], y[0] == 5}, y[t], t];

Plot[Evaluate[y[t] /. sol], {t, 0, 10},
 PlotLabel → "Exponential Growth",
 AxesLabel → {"t", "y"}]
```

Out[◦]=

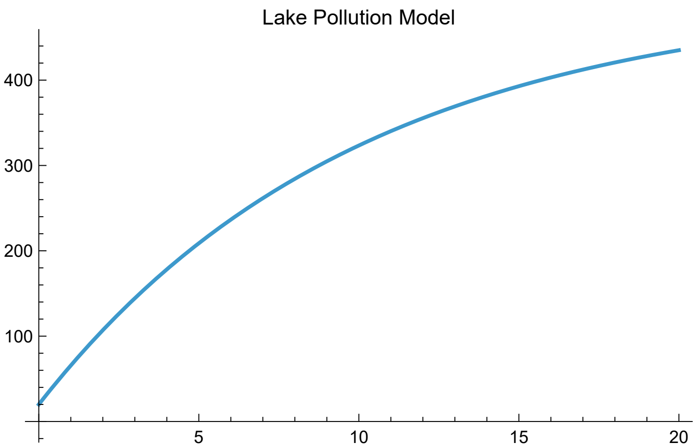


In[1]:=
```
qin = 10; qout = 10; cin = 5;

sol = DSolve[{x'[t] == qin cin - qout x[t] / 100, x[0] == 20}, x[t], t];

Plot[Evaluate[x[t] /. sol], {t, 0, 20},
 PlotLabel → "Lake Pollution Model"]
```

Out[3]=

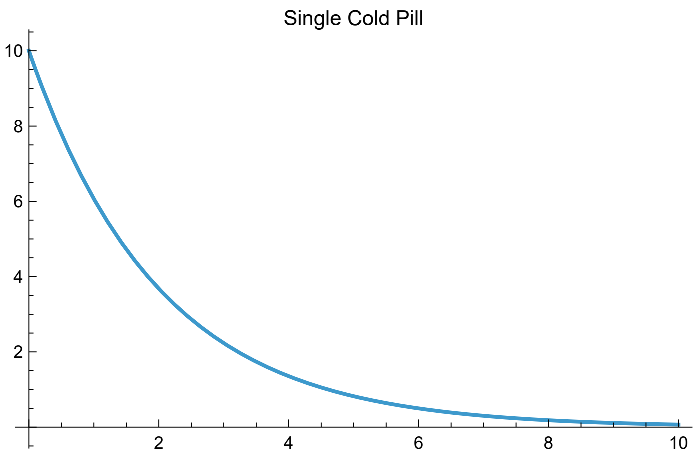


In[∘]:=
```
k = 0.5;

sol = DSolve[{c'[t] == -k c[t], c[0] == 10}, c[t], t];

Plot[Evaluate[c[t] /. sol], {t, 0, 10},
 PlotLabel → "Single Cold Pill"]
```

Out[∘]=



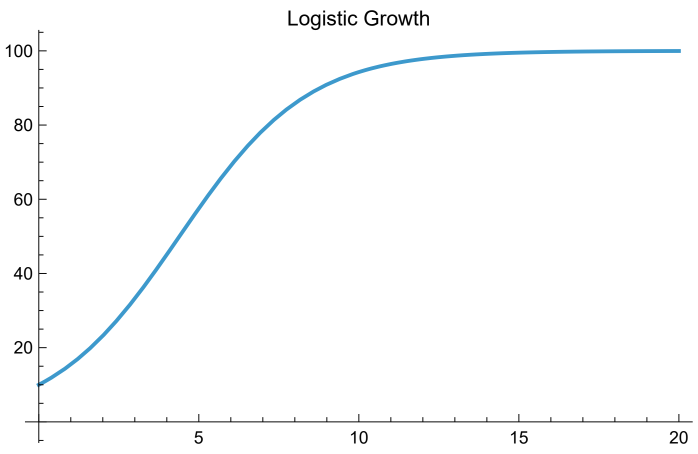
In[∘]:=
```
r = 0.5; K = 100;

sol = DSolve[{p'[t] == r p[t] (1 - p[t] / K), p[0] == 10}, p[t], t];

Plot[Evaluate[p[t] /. sol], {t, 0, 20},
 PlotLabel → "Logistic Growth"]
```

DSolve: Inverse functions are being used by DSolve, so some solutions may not be found.

Out[∘]=

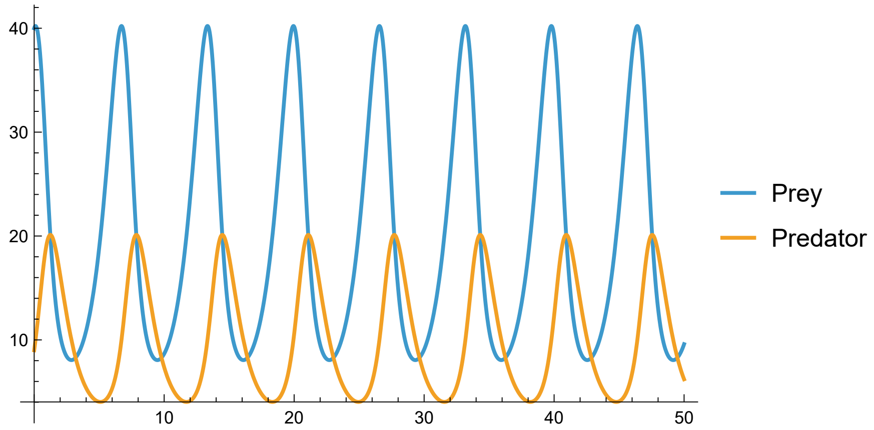


In[∘]:=

In[•]:=
```
sol = NDSolve[
  {
  x'[t] == x[t] (1 - 0.1 y[t]),
  y'[t] == -y[t] + 0.05 x[t] × y[t],
  x[0] == 40, y[0] == 9
  },
  {x, y}, {t, 0, 50}
 ];

Plot[Evaluate[{x[t], y[t]} /. sol], {t, 0, 50},
 PlotLegends → {"Prey", "Predator"}]
```

Out[•]=



In[∘]:=
```
sol = NDSolve[
 {
 s'[t] == -0.002 s[t] × i[t],
 i'[t] == 0.002 s[t] × i[t] - 0.5 i[t],
 r'[t] == 0.5 i[t],
 s[0] == 990, i[0] == 10, r[0] == 0
 },
 {s, i, r}, {t, 0, 50}
];

Plot[Evaluate[{s[t], i[t], r[t]} /. sol], {t, 0, 50},
 PlotLegends → {"Susceptible", "Infected", "Recovered"}]
```

NDSolve: The equations {0.5[0] == 0} are not differential equations or initial conditions in the dependent variables {i, s}.

ReplaceAll:
{NDSolve[{s′[t] == −0.002 i[t] s[t], i′[t] == −0.5 i[t] + 0.002 i[t] s[t], 0 == 0.5 i[t], s[0] == 990, i[0] == 10, 0.5[0] == 0}, {s, i, 0.5}, {t, 0, 50}]}
is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

NDSolve: 0.0010214285714285716` cannot be used as a variable.

ReplaceAll:
{NDSolve[{s′[0.00102143] == −0.002 i[0.00102143] s[0.00102143], i′[0.00102143] == −0.5 i[0.00102143] + 0.002 i[0.00102143] s[0.00102143], 0 == 0.5 i[0.00102143], s[0] == 990, i[0] == 10, 0.5[0] == 0}, {s, i, 0.5}, {0.00102143, 0, 50}]}
is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

NDSolve: 0.0010214285714285716` cannot be used as a variable.

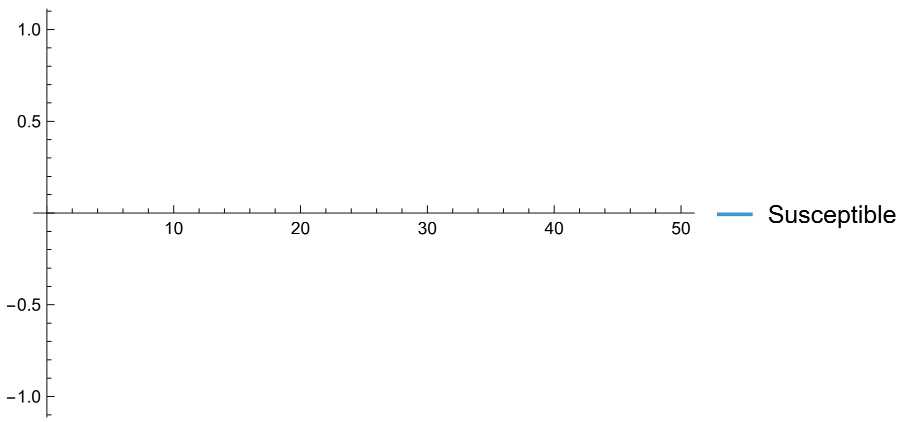
ReplaceAll:
{NDSolve[{s′[0.00102143] == −0.002 i[0.00102143] s[0.00102143], i′[0.00102143] == −0.5 i[0.00102143] + 0.002 i[0.00102143] s[0.00102143], 0. == 0.5 i[0.00102143], s[0.] == 990., i[0.] == 10., 0.5[0.] == 0.}, {s, i, 0.5}, {0.00102143, 0., 50.}]}
is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

General: Further output of ReplaceAll::reps will be suppressed during this calculation.

NDSolve: 1.0214295918367347` cannot be used as a variable.

General: Further output of NDSolve::dsvar will be suppressed during this calculation.

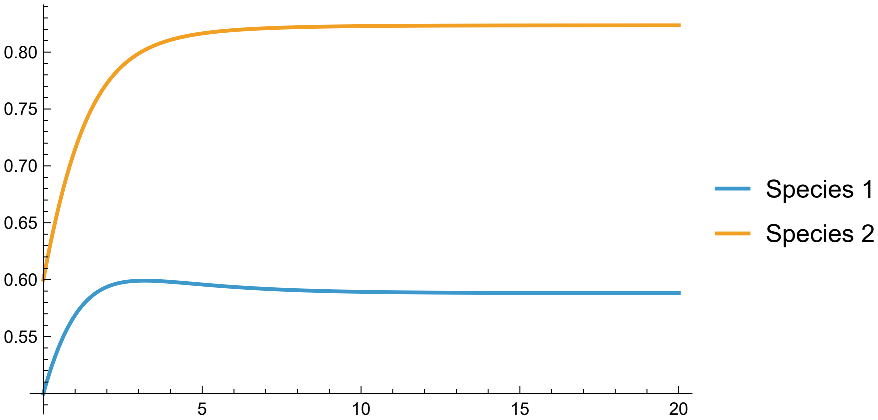
Out[∘]=



In[ ]:=
```
sol = NDSolve[
 {
 x'[t] == x[t] (1 - x[t] - 0.5 y[t]),
 y'[t] == y[t] (1 - y[t] - 0.3 x[t]),
 x[0] == 0.5, y[0] == 0.6
 },
 {x, y}, {t, 0, 20}
];

Plot[Evaluate[{x[t], y[t]} /. sol], {t, 0, 20},
 PlotLegends → {"Species 1", "Species 2"}]
```

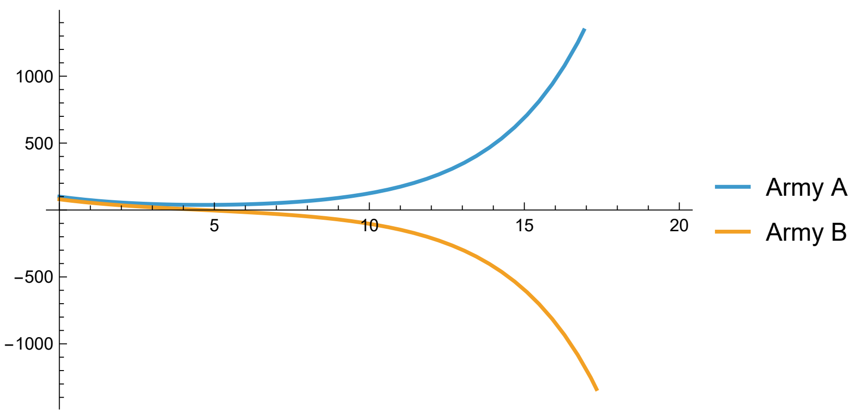
Out[ ]=



In[ ]:=
```
sol = NDSolve[
 {
 x'[t] == -0.4 y[t],
 y'[t] == -0.3 x[t],
 x[0] == 100, y[0] == 80
 },
 {x, y}, {t, 0, 20}
];

Plot[Evaluate[{x[t], y[t]} /. sol], {t, 0, 20},
 PlotLegends → {"Army A", "Army B"}]
```

Out[ ]=



```
In[•]:= pts = RandomReal[{0, 1}, {10000, 2}];

inside = Count[pts, {x_, y_} /; y ≤ x^2];

area = inside / 10000
```

Out[•]=

$$\frac{1737}{5000}$$

```
data = {{1, 2}, {2, 5}, {3, 10}, {4, 17}};

model = Fit[data, {1, x, x^2}, x]
```

Out[•]=

$1. + 2.83536 \times 10^{-15}\, x + 1.\, x^2$

```
In[•]:= data = {{1, 2}, {2, 8}, {3, 18}, {4, 32}};

FindFit[data, a x^n, {a, n}, x]
```

Out[•]=

{a → 2., n → 2.}

```
In[•]:= FindFit[data, b x^n, {b, n}, x]
FindFit[data, b Exp[a x], {a, b}, x]
FindFit[data, a Log[x] + b, {a, b}, x]
FindFit[data, a x^2, {a}, x]
FindFit[data, a x^3, {a}, x]
```

Out[•]=

{b → 1.52769, n → 1.73248}

Out[•]=

{a → 0.61496, b → 1.4732}

Out[•]=

{a → 10.1506, b → 0.435176}

Out[•]=

{a → 1.08475}

Out[•]=

{a → 0.286299}